Worst-case to Average-case Hardness of LWE:

A Simple and Practical Perspective

Alexandra Veliche

joint with Divesh Aggarwal & Leong Jin Ming

Roadmap

I. LWE Problem and Why it is Important

II. Alternative Perspective on Computational Hardness

III. Our Results

IV. Proof Techniques

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Learning With Errors

On Lattices, Learning with Errors, Random Linear Codes, and Cryptography

Oded Regev *

May 2, 2009

Abstract

Our main result is a reduction from worst-case lattice problems such as GAPSVP and SIVP to a certain learning problem. This learning problem is a natural extension of the 'learning from parity with error' problem to higher moduli. It can also be viewed as the problem of decoding from a random linear code. This, we believe, gives a strong indication that these problems are hard. Our reduction, however, is quantum. Hence, an efficient solution to the learning problem implies a *quantum* algorithm for GAPSVP and SIVP. A main open question is whether this reduction can be made classical (i.e., non-quantum).

We also present a (classical) public-key cryptosystem whose security is based on the hardness of the learning problem. By the main result, its security is also based on the worst-case quantum hardness of GAPSVP and SIVP. The new cryptosystem is much more efficient than previous lattice-based cryptosystems: the public key is of size $\tilde{O}(n^2)$ and encrypting a message increases its size by a factor of $\tilde{O}(n)$ (in previous cryptosystems these values are $\tilde{O}(n^4)$ and $\tilde{O}(n^2)$, respectively). In fact, under the assumption that all parties share a random bit string of length $\tilde{O}(n^2)$, the size of the public key can be reduced to $\tilde{O}(n)$.

Learning With Errors

[Regev, 2009] – reduction from worst-case lattice problems to decision-LWE

$$\boxed{\text{GapSVP}} \xrightarrow{\text{quantum}} \boxed{\text{BDD}} \xrightarrow{\text{classical}} \boxed{\text{LWE}}$$

Learning With Errors

[Peikert, 2009], [Brakerski+,2013] – fully classical reduction that preserves parameters

$$\boxed{\text{GapSVP}} \xrightarrow{\text{classical}} \boxed{\text{BDD}} \xrightarrow{\text{classical}} \boxed{\text{LWE}}$$

Learning With Errors

Worst-case hardness of standard lattice problems $\Longrightarrow$ Average-case hardness $\Longrightarrow$ Lattice-based cryptosystems

$\boxed{\text{GapSVP}}$ $\boxed{\text{BDD}}$ $\boxed{\text{LWE}}$

Learning With Errors

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$\boxed{\text{GapSVP}}$ $\boxed{\text{BDD}}$ $\boxed{\text{LWE}}$

Post-quantum cryptography!

no known efficient quantum algorithms for these problems

Learning With Errors

$\mathbf{LWE}_{n,p,\phi}$: n = dimension, p = modulus, ϕ = error distribution over $\mathbb{R}/\mathbb{Z}$

Given n^k noisy samples of the form $\left(\mathbf{a}, b = \frac{1}{p}\langle \mathbf{a}, \mathbf{s}\rangle + e\right)$, where

$\mathbf{a} \leftarrow \mathbb{Z}_p^n$ uniformly random, $e \leftarrow \phi$ error, and $\mathbf{s} \in \mathbb{Z}_p^n$ unknown secret vector,

(if samples are uniformly random over $\mathbb{Z}_p^n \times \mathbb{R}/\mathbb{Z}$.

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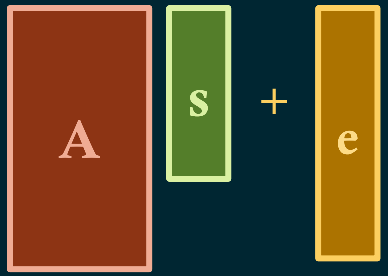

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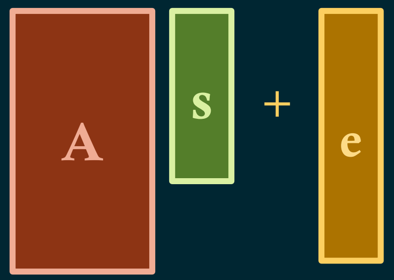
(search-LWE) output $\mathbf{s}$.

(decision-LWE) output

- YES if samples are from the $\mathbf{LWE}$ distribution determined by $\mathbf{s}$ and ϕ,
- NO if samples are uniformly random over $\mathbb{Z}_p^n \times \mathbb{R}/\mathbb{Z}$.



Algorithms for LWE

[Blum-Kalai-Wasserman, 2000] – Best known algorithm for $\mathrm{LWE}_{n,p,\phi}$ runs in $p^{O(n/\log n)}$ time.

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$\gamma = \mathrm{poly}(n)$

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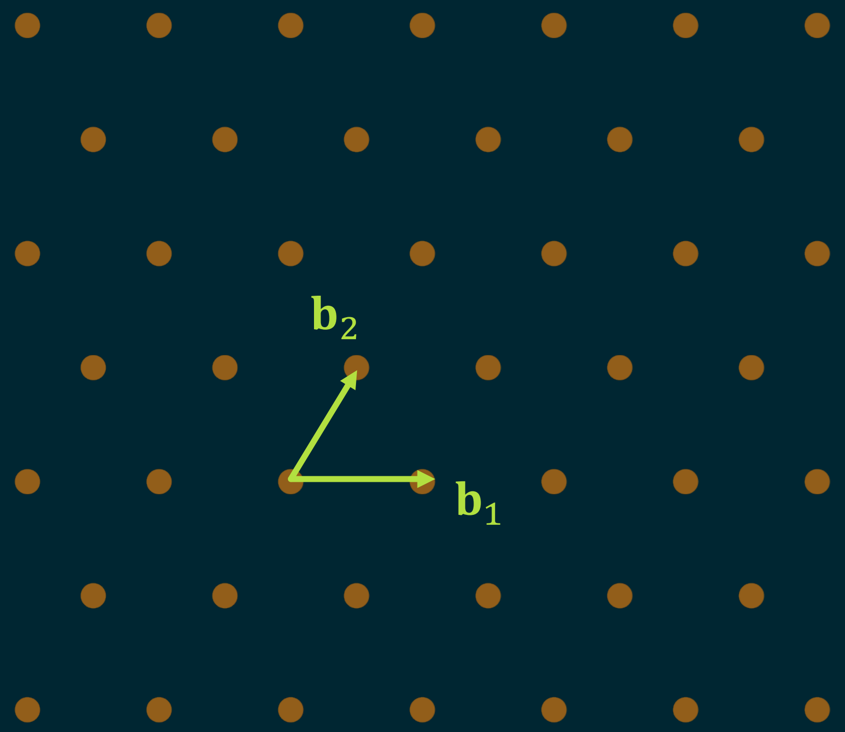
Gap!

Lattices

Lattice: A discrete additive subgroup of $\mathbb{R}^n$ consisting of all linear combinations of linearly independent basis vectors $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_\mathrm{n}\} \subset \mathbb{R}^n$:

$$\mathcal{L} = \mathcal{L}(\mathcal{B}) = \left\{ \sum_{i=1}^{n} a_i \mathbf{b}_i : a_1, \dots, a_n \in \mathbb{Z} \right\}.$$



Shortest Vector Problem

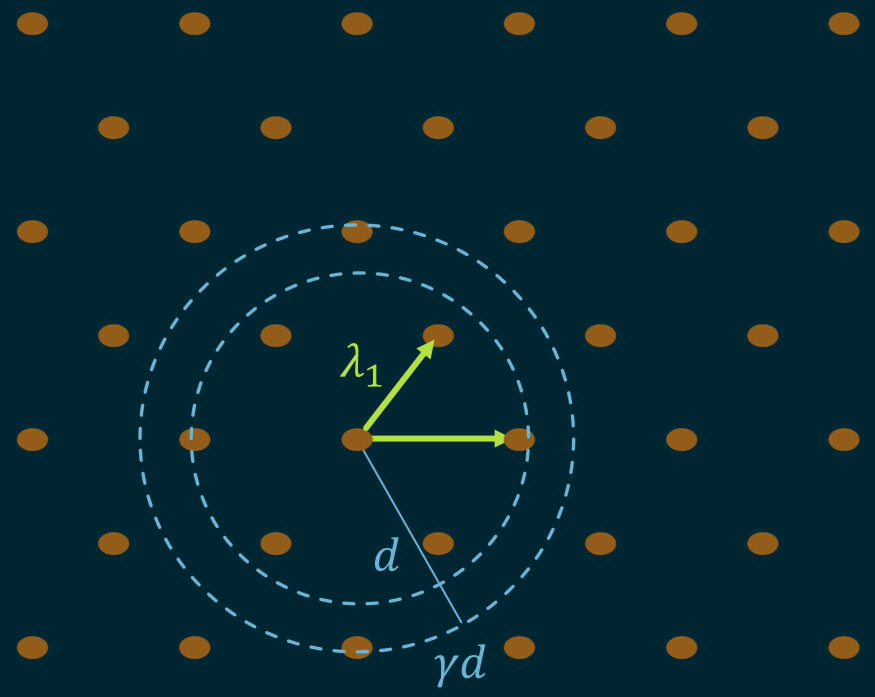
GapSVP$_\gamma$: $\gamma \geq 1$ approximation factor

Given a basis $\mathcal{B}$ for a full-rank lattice $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}}(\mathcal{B}) \subseteq \mathbb{R}^n$

and a distance parameter $d > 0$,

output

- YES if $\lambda_1(\boldsymbol{\mathcal{L}}) \leq d$
- NO if $\lambda_1(\boldsymbol{\mathcal{L}}) \geq \gamma \cdot d$.



Bounded Distance Decoding

$\mathbf{BDD}_\alpha$: $\alpha =$ approximation factor

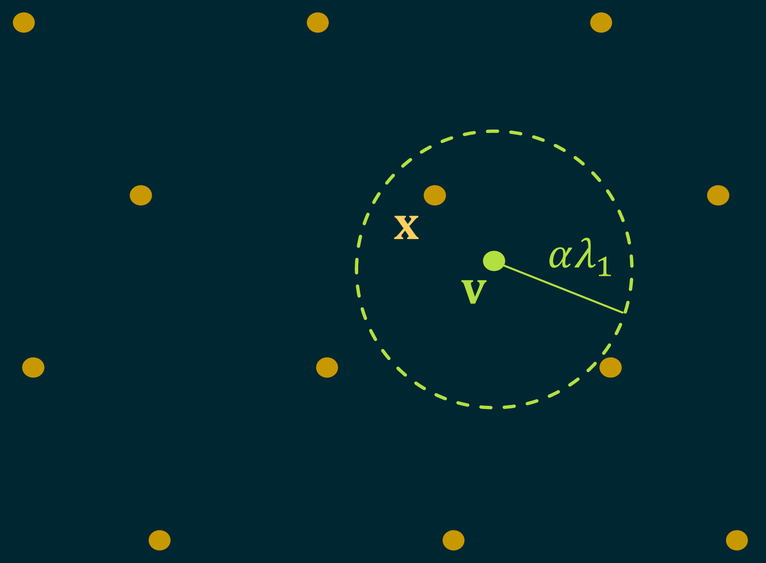
Given a basis $\mathcal{B}$ for a full-rank lattice $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}}(\mathcal{B}) \subseteq \mathbb{R}^n$

and a target vector $\mathbf{v} \in \mathbb{R}^n$,

with the promise that $dist(\mathbf{v}, \mathcal{L}) < \alpha \cdot \lambda_1(\mathcal{L})$,

find a lattice vector closest to $\mathbf{v}$,

i.e. $\mathbf{x} \in \mathcal{L}$ such that $\|\mathbf{v} - \mathbf{x}\| < \alpha \cdot \lambda_1(\mathcal{L})$.



Bounded Distance Decoding

BDD$_{\alpha}$: $\alpha \in \left(0, \frac{1}{2}\right)$ approximation factor

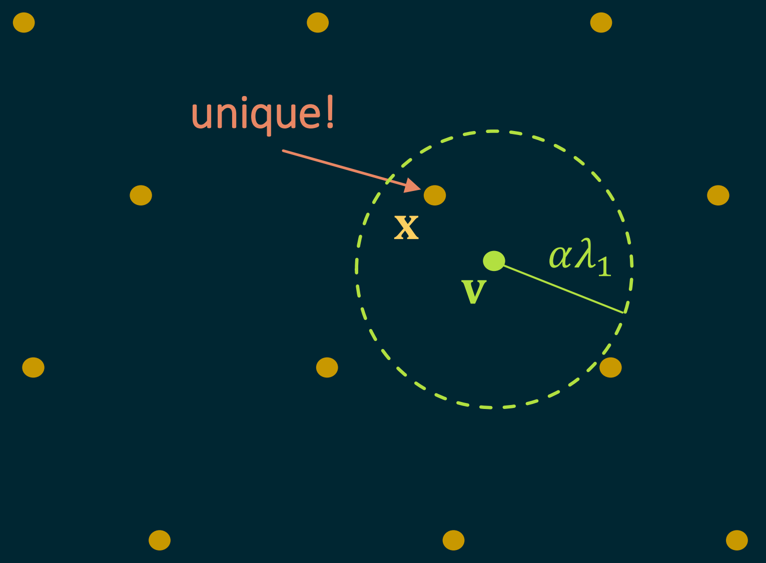
Given a basis $\mathcal{B}$ for a full-rank lattice $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}}(\mathcal{B}) \subseteq \mathbb{R}^n$

and a target vector $\mathbf{v} \in \mathbb{R}^n$,

with the promise that $dist(\mathbf{v}, \mathcal{L}) < \alpha \cdot \lambda_1(\mathcal{L})$,

find *the unique* lattice vector closest to $\mathbf{v}$,

i.e. $\mathbf{x} \in \mathcal{L}$ such that $\|\mathbf{v} - \mathbf{x}\| < \alpha \cdot \lambda_1(\mathcal{L})$.



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Bit Security

What does it mean for a cryptosystem to be 256-bit secure?

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What does it mean for a cryptosystem to be 256-bit secure?

(a) The fastest algorithm for breaking the cryptosystem runs in 2^{256} time.

(b) An efficient algorithm succeeds in attacking the cryptosystem with probability $> 2^{-256}$.

Complexity Framework

[Aggarwal-Maurer, 2011] – *unpredictability entropy* of a computational problem $\mathcal{P}$:

If p is the maximum success probability of any PPT algorithm that solves $\mathcal{P}$,

then its unpredictability entropy is $\log_2 p$.

Complexity Framework

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Complexity Framework

Reduction algorithm for $\mathcal{P} \rightarrow \mathcal{Q}$ makes k calls to oracle for $\mathcal{Q}$.

Success probability of solving $\mathcal{P}$ is $\delta \implies$ success probability of solving $\mathcal{Q}$ is $\delta^{1/k}$.

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Use only one-shot reductions!

Reduction from Lattice Problems

Success probability of any PPT algorithm for $\mathrm{LWE}_{n,p}$ is p^{-n}.

$$\boxed{\mathrm{GapSVP}_\gamma} \xrightarrow{\text{classical}} \boxed{\mathrm{BDD}_\alpha} \xrightarrow{\text{classical}} \boxed{\mathrm{LWE}_{n^2,p,\phi}}$$

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$\gamma = \mathrm{poly}(n)$

$p = \mathrm{poly}(n)$

Success probability of best known PPT algorithm for $\mathrm{GapSVP}_{n,\gamma}$ is $2^{-\Theta(n^2/\log n)}$.

Complexity Framework

Conjecture:

For any constants $c, c' > 0$, there is exists $\kappa = \kappa(c, c') > 0$ such that

no algorithm can solve BDD_α on an arbitrary n-rank lattice for $\alpha = \frac{1}{n^c}$ in time $n^{c'}$

with success probability better than $2^{-n^2/\kappa\ \log n}$.

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Assuming this conjecture, can we show a lower bound of $p^{-\Omega(n)}$

on the success probability of any PPT algorithm that solves $\mathrm{LWE}_{n,p}$?

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Our Results

Theorem 1: *(informal)* If no PPT algorithm can solve BDD_α for approximation factor $\alpha \in \left(0, \frac{1}{2}\right)$

with success probability greater than $2^{-\frac{n^2}{\kappa \log n} - 4}$ for some $\kappa > 0$,

then no PPT algorithm can solve search-$\mathrm{LWE}_{n,p,\phi}$ with binary secret, dimension n, and

modulus $p = \mathrm{poly}(n)$ with success probability $2^{-\frac{n^2}{\kappa \log n}}$.

Main Problem

[Aggarwal-Maurer, 2011] – *unpredictability entropy* of a computational problem $\mathcal{P}$:

If p is the maximum success probability of any PPT algorithm that solves $\mathcal{P}$,

then its unpredictability entropy is $\log_2 p$.

What does this mean for decision problems?

Learning With Errors

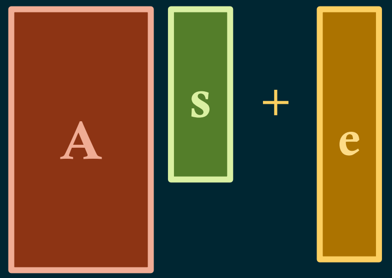
decision-LWE$_{n,p,\phi}$: n = dimension, p = modulus, ϕ = error distribution over $\mathbb{R}/\mathbb{Z}$

Given n^k noisy samples of the form $\left(\mathbf{a}, \frac{1}{p}\langle \mathbf{a}, \mathbf{s} \rangle + e\right)$, where $\mathbf{a} \leftarrow \mathbb{Z}_p^n$ uniformly random,

$e \leftarrow \phi$ error, and $\mathbf{s} \in \mathbb{Z}_p^n$ unknown secret vector,

output

- YES if samples are from the LWE distribution determined by $\mathbf{s}$ and ϕ,
- NO if samples are *uniformly random* over $\mathbb{Z}_p^n \times \mathbb{R}/\mathbb{Z}$.



Complexity Framework

If there is no PPT algorithm that solves search-$\mathrm{LWE}_{n,p}$ with success probability α,

then can we show that there is no algorithm that solves decision-$\mathrm{LWE}_{n,p}$ and outputs

- YES with probability $\alpha' \approx \alpha$ given a YES instance
- YES with probability $\beta' \ll \alpha'$ given a NO instance?

Complexity Framework

If there is no PPT algorithm that solves search-$\mathrm{LWE}_{n,p}$ with success probability α,

then can we show that there is no algorithm that solves decision-$\mathrm{LWE}_{n,p}$ and answers

- Correctly with probability $\approx \alpha$
- Incorrectly with probability $\beta \approx 0$
- $\perp$ with probability $\approx 1 - \alpha$?

Our Results

Theorem 2: *(informal)* If no algorithm can solve search-$\mathrm{LWE}_{n,p}$ for modulus $p = \mathrm{poly}(n)$

with success probability α in *expected* polynomial time,

then no PPT algorithm can solve decision-$\mathrm{LWE}_{n,p}$ for modulus p that

- outputs the correct answer with probability α and
- outputs $\perp$ with probability $1 - \alpha$.

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Chain of Reductions

I.

$\boxed{\mathrm{BDD}_{\alpha}} \longrightarrow \boxed{\text{mod-BDD}_{\alpha,p}} \longrightarrow \boxed{\text{gen-LWE}_{n,p,\mathcal{D}}} \longrightarrow \boxed{\mathrm{LWE}_{n^2,p}}$

Modular Bounded Distance Decoding

mod-BDD$_{\alpha,p}$: α = approximation factor, p = modulus

Given a basis $\mathcal{B}$ for a full-rank lattice $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}}(\mathcal{B}) \subseteq \mathbb{R}^n$

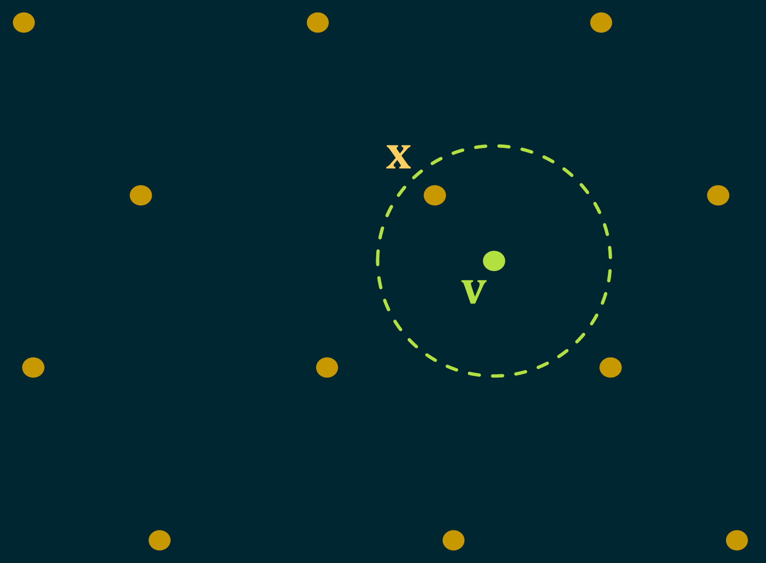
and a target vector $\mathbf{v} \in \mathbb{R}^n$,

with the promise that $dist(\mathbf{v}, \mathcal{L}) < \alpha \cdot \lambda_1(\mathcal{L})$,

find the coefficient vector of a lattice vector closest to $\mathbf{v} \bmod p$,

i.e. $\mathbf{B}^{-1}\mathbf{x} \bmod p \in \mathbb{Z}_p^n$ for some

$\mathbf{x} \in \mathcal{L}$ such that $\|\mathbf{v} - \mathbf{x}\| < \alpha \cdot \lambda_1(\mathcal{L})$.



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Blow up the modulus!
use $p = 2^n$

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Blow up the modulus!
use $p = 2^n$

success prob. q $\longleftarrow$ success prob. q

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Generate discrete Gaussian
samples and use them to
generate LWE samples

Chain of Reductions

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success prob. $q - \epsilon$ $\longleftarrow$ success prob. q

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Carefully sample Gaussian noise that guarantees optimal success probability

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Carefully sample Gaussian noise that guarantees optimal success probability

success prob. $\frac{q}{(1+\epsilon)^3}$ $\longleftarrow$ success prob. q

Chain of Reductions

I.

$$\begin{tikzcd}
\boxed{\mathrm{BDD}_{\alpha}} \arrow[r] & \begin{array}{c}\boxed{\text{mod-BDD}_{\alpha,p}}\\ \text{modulus}\\ p = 2^n\end{array} \arrow[r] & \begin{array}{c}\boxed{\text{gen-LWE}_{n,p,\mathcal{D}}}\\ \text{modulus}\\ p = 2^n\end{array} \arrow[r] & \begin{array}{c}\boxed{\mathrm{LWE}_{n,p,\phi}}\\ \text{modulus}\\ p = 2^n\end{array} \arrow[d] \\
 & & \begin{array}{c}\boxed{\mathrm{LWE}_{n^2,p,\phi}}\\ \text{modulus}\\ p = n^{\Theta(1)}\end{array} & \begin{array}{c}\boxed{\text{binary-LWE}_{n^2,p,\phi}}\\ \text{modulus}\\ p = n^{\Theta(1)}\end{array} \arrow[l]
\end{tikzcd}$$

Chain of Reductions

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\end{tikzcd}$$

$\text{mod-BDD}_{\alpha,p}$: modulus $p = 2^n$

$\text{gen-LWE}_{n,p,\mathcal{D}}$: modulus $p = 2^n$

$\mathrm{LWE}_{n,p,\phi}$: modulus $p = 2^n$, success prob.q

$\text{binary-LWE}_{n^2,p,\phi}$: modulus $p = n^{\Theta(1)}$, success prob.q

$\mathrm{LWE}_{n^2,p,\phi}$: modulus $p = n^{\Theta(1)}$, success prob.q

Chain of Reductions

II.

$\boxed{\text{search-LWE}_{n^2,p,\phi}} \longrightarrow \boxed{\text{GL-LWE}_{n,p,\phi}} \longrightarrow \boxed{\text{decision-LWE}_{n,p,\phi}}$

Goldreich-Levin LWE

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and a $\mathbf{r} \in \mathbb{Z}_p^n$ uniformly random vector,

output $\langle\mathbf{s},\mathbf{r}\rangle \bmod p$.

Goldreich-Levin Theorem

Goldreich-Levin Theorem: *(informal)*

Given any one-way function f, the function h defined by $h(\mathrm{x}, \mathrm{r}) = \langle \mathrm{x}, \mathrm{r} \rangle \bmod 2$

is a hard-core predicate for the function g defined by $g(\mathrm{x}, \mathrm{r}) = (f(\mathrm{x}), \mathrm{r})$.

Goldreich-Levin Theorem

Levin's Theorem: *(informal)*

Given any one-way function f, the function h defined by $h(\mathrm{x},\mathrm{r}) = \langle \mathrm{x},\mathrm{r}\rangle \bmod 2$

is a hard-core predicate for the function g defined by $g(\mathrm{x},\mathrm{r}) = (f(\mathrm{x}),\mathrm{r})$.

The maximum success probability of finding $h(\mathrm{x},\mathrm{r})$ given $f(\mathrm{x}),\mathrm{r}$ is

the maximum success probability of finding x given $f(\mathrm{x})$.

Goldreich-Levin Theorem

Theorem: *(generalization)*

Let $p > 10.$ Given any one-way function $f,$ the function h defined by $h(\mathrm{x}, \mathrm{r}) = \langle \mathrm{x}, \mathrm{r} \rangle \bmod p$ is a hard-core predicate for the function g defined by $g(\mathrm{x}, \mathrm{r}) = (f(\mathrm{x}), \mathrm{r})$.

The maximum success probability of finding $h(\mathrm{x}, \mathrm{r})$ given $f(\mathrm{x}), \mathrm{r}$ is the maximum success probability of finding x given $f(\mathrm{x})$.

Chain of Reductions

II.

$$\text{search-LWE}_{n^2,p,\phi} \longrightarrow \text{GL-LWE}_{n,p,\phi} \longrightarrow \text{decision-LWE}_{n,p,\phi}$$

Use the GL reduction!

Chain of Reductions

II.

search-LWE$_{n^2,p}$ $\longrightarrow$ GL-LWE$_{n,p}$ $\longrightarrow$ decision-LWE$_{n,p}$

Re-randomize samples
and run oracle on them

Future Directions

- The two chains of reductions are disconnected, because *expected* polynomial-time reduction is a fundamental part of the GL theorem. Is there a workaround?
- Establish a similar result for GapSVP to complete the chain of reductions to BDD.
- Use this alternative framework to study the complexity of other computational problems relevant to cryptography or learning problems.

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Thank you! Questions?